

SEG-RobustEye: Understanding Medical Image Segmentation Models

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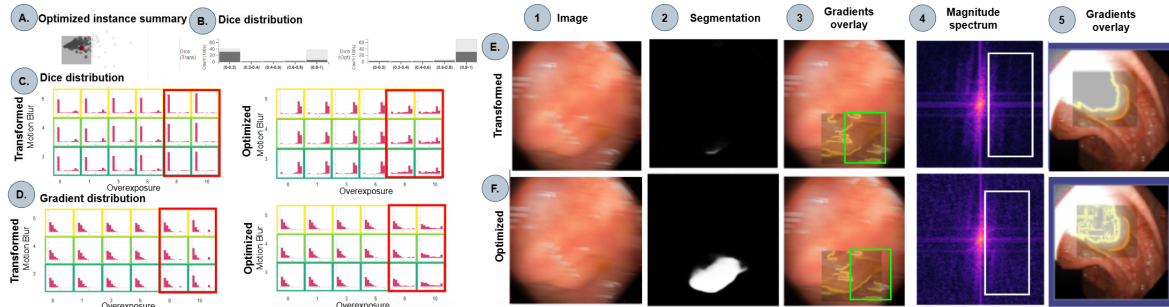


Figure 1: SEG-RobustEye a system for robustness analysis of medical segmentation models under input transformations. (A) Optimized instance summary to explore global behavior per instance. Outliers can be filtered by user selection. (B) Dice distribution over the whole dataset. (C) Dice distribution and (D) Gradient distribution for the selected subgroup. Transformed (E) vs Optimized (F) for an instance. (1) Input image. (2) Segmentation output. (3) Gradient overlays. (4) Magnitude spectrum. (5) Gradient pattern added after optimization in overexposed areas.

ABSTRACT

Deep learning (DL) models have proven to be suitable for various applications, achieving state-of-the-art performance. Despite that, they experience notable performance drops when subjected to realistic transformations in input data. Analyzing model behavior under input transformations is essential to preemptively identify possible breaking points and understand what image characteristics might cause them before their failure in the real world.

We introduce SEG-RobustEye, a visual analytics (VA) design developed to assist in the evaluation of the robustness of segmentation models for endoscopy images under realistic input transformations. SEG-RobustEye is based on ProactiV [13], a VA framework designed for understanding the behaviour of image-to-image translation models. SEG-RobustEye is a tailored instance of the framework and an extension of ProactiV for medical images, in concrete endoscopic images. These require visual designs that enhance the relevant features for such medical applications, which are different from general natural scenes. SEG-RobustEye is designed to discover features that affect the model behaviour under specific transformations. SEG-RobustEye connects the provided perspectives by ProactiV, i.e., global and instance level, and extends to subgroup level patterns. Subgroup level patterns facilitate the discovery and generalizability of selected subgroups of instances. The value of our approach was verified against real-world cases in endoscopy imaging by DL developers as proof of concept of the potential of SEG-RobustEye and, by extension, of ProactiV.

Index Terms: Visual analytics, explainable AI, medical imaging, input transformations, robustness analysis, model behavior, deep learning

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1 INTRODUCTION

Endoscopy is one of the main screening medical procedures used to visualize the gastrointestinal tract. It involves the insertion of a high-resolution camera (i.e., endoscope) into the body. During an endoscopy, healthcare professionals can identify abnormalities like inflammation, ulcers, tumors, or bleeding. Endoscopic procedures are prone to clinical errors. Recently, the focus has shifted towards computer-aided diagnosis systems to automate this process, where DL models are used for polyps detection and segmentation.

Although these models achieve state-of-the-art performance [9], an unnoticeable change in the original image can prove to have adverse and random effects on the output, and lead to undesired model behavior [6][9]. For example, a compressed image can look the same as the original, however, the output of the model can shift from a fully segmented polyp to no detection [6]. Understanding the behavior of the model is of high importance for deployment. Furthermore, multiple input transformations are present at the same time, adding complexity to the model analysis [6]. The semantically relevant features of medical images are subtle and often not visible to the naked eye, or naive representation. The effect of input transformations on the DL models was studied before [8][1][2][4][12], particularly for endoscopy [6][5]. Previous studies focus on predefining the types of input transformations that might be relevant for the specific application. In most cases, the exploration is limited to independent transformations, not including their combinations. Furthermore, the studies on medical imaging are limited to analyzing performance metrics (i.e., if a model fails) or studying the effect on a few instances, not allowing a holistic analysis. This analysis is not able to provide insight into why, meaning what specific features determine the model failure. Since understanding model behavior is exploratory in nature, including a human in the loop is a suitable approach.

ProactiV [13] proposes a framework for models for image-to-image transformation. However, ProactiV is not adapted for the needs of analyzing semantically complex medical data, such as endoscopy images, and it is not designed for segmentation tasks. We propose SEG-RobustEye, an instance and extension to ProactiV

[13] with the following contributions. First, it adapts the framework and optimization method to the segmentation task. The input is optimized to project transformed instances into a space that yields desired segmentation output (i.e., ground truth). This allows to identify what changes are needed to the original image to achieve the desired segmentation. Second, it approaches the semantical complexity of medical images. Relevant image features are identified, and their visual representation is designed with the involvement of domain experts, i.e., machine learning experts. Third, it provides a subgroup of instances analysis component to the framework. ProactiV provides instance and global level (i.e., all instances), subgroup analysis generalizes the individual instance-level providing a smoother transition between the local and global levels of analysis.

2 RELATED WORK AND BACKGROUND

2.1 Visual Analytics for Model Behavior

Research has shown that using VA significantly improves model interpretability. Numerous systems focus on exploring the behavior of a DL model, primarily in the context of classification tasks. Research on methods for interpreting deep neural networks is common, for example, based on the visualization of surrogate models [16]. Another perspective is exploring model behavior by identifying areas of improvement for classification tasks [11] using data embeddings to discover instances of interest, and data augmentation to explore how the model output changes. Some works propose model behavior understanding through analysis of the data distribution and how models react under input perturbations. For example, ScatterUQ [10] allows prediction interpretation through in-distribution and out-of-distribution analysis, and Perturber [15] analyses activations and predictions in CNNs under input perturbations. To the best of our knowledge, ProactiV [13] is the only method that provides a complete and interactive analysis from global to instance level of the model behavior and breaking points under input transformations.

None of the methods above are adequate for medical data such as endoscopic images. Moreover, existing visual analytics systems that approach medical data usually explore data distributions [3], use data-specific characteristics to explore model results [14], or only look at the model certainty of the output [17]. While ProactiV explores model behavior under input transformations for models dealing with complex outputs beyond classification, it falls short of capturing the semantical complexity of medical images and is not trivially adapted to segmentation tasks.

2.2 ProactiV

The ProactiV [13] design includes two key methods: summarization, which provides a holistic view of model behavior across individual data instances and transformations, and input optimization, which identifies input-output patterns and performance issues.

The input optimization method examines how input transformations affect model behavior by projecting transformed instances into a space where the model performs correctly. The assumption is that the differences between the transformed input and the optimized one indicate what the model expects to be changed to provide the desired output. Transformed instances (e.g. increasing

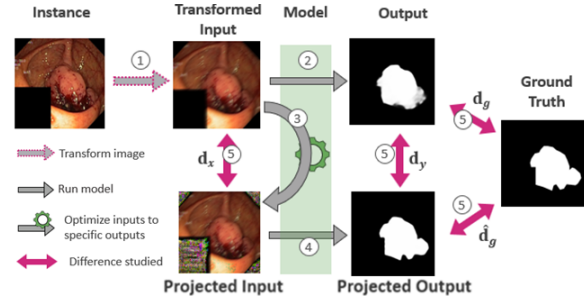


Figure 2: Adaptation of optimization strategy of ProactiV Figure 1 [13]

noise) potentially lead to incorrect outputs for some transformations. Input optimization adjusts the transformed input using the model’s loss function while keeping model weights fixed, generating a “Projected input” (see Fig. 2.3). By comparing the original and projected outputs with the ground truth, insights into model robustness and stability under different transformations can be gained.

ProactiV provides a method to analyze model behavior at a global level by summarizing differences over transformations and instances, by using distance vectors to represent the transformations in input and output. These distance vectors are used to create summaries of the model’s response to various transformations or over instances, which can then be summarized using dimensionality reduction techniques to reveal patterns or trends in model behavior across different transformations or instances. We adapt these methods from ProactiV by using the dice score, a common metric for segmentation to measure differences between outputs, and for model validation [9]. To quantify differences between input images and optimized in the global and subgroup views, we use MSE, KL-divergence for the color characteristics, and edge intensity.

2.3 Dataset and Transformations

In this section, we present the data, model and relevant transformations that will be used through the paper. We use a public dataset Kvasir-SEG [7] that contains high-resolution endoscopy images and the expected polyp segmentation masks. As transformations that are interesting to analyze by the users, we use the setup defined in the study by Jaspers et al. [6]. It includes eleven input transformations organized in three different expert-defined categories, commonly present for endoscopic imaging: postprocessing, user-dependent, and compression-induced input transformations. As model to be analyzed, we use the one presented by Jaspers et al. [6]. The model was built using the U-Net architecture, with an EfficientNet-B6 backbone as the encoder as described by Kunsters et al. and trained by domain experts [9]. This model was trained on the Kvasir-SEG dataset, with an average dice score of 0.83, which is considered to be close to state-of-the-art performance on the Kvasir-SEG test set. The model is trained by domain experts on Kvasir-SEG dataset. The data was split before training in 50 % train, 25 % validation, 25 % test. For our analysis, only the test dataset was used.

3 DESIGN PROCESS AND TASKS

In endoscopy, input corruptions (i.e., undesired transformations) are common and inevitable due to the nature of the image generation process. While user-dependent transformations could depend on the skills of the clinician, input transformations such as compression-induced or postprocessing cannot be avoided.

Since we use ProactiV [13] as the basis, the ProactiV defined tasks are also applicable. Feedback from deep learning experts working on developing models for endoscopic images segmentation was gathered through multiple sessions during the design process. First, there was a session with one senior expert, and a second

session where 8 junior researchers were involved. The challenges in model building were discussed in both sessions. VA for model analysis in general and ProactiV specifically were presented and discussed. Regular meetings (i.e., once each two weeks during nine months) were also arranged with one of the junior researchers to discuss the design developments. During the first two sessions, the ProactiV tasks were verified. We also identified the need of an intermediate component between global and independent instance level analysis, i.e., subgroup analysis component of selected groups of instances. This created two main tasks: identification of relevant subgroups and discovering patterns at subgroup level.

4 DESIGN AND VISUALIZATION METHODS

Fig. 1 illustrates some of the extensions of SEG-RobustEye from the base design of ProactiV [13]. ProactiV facilitates the analysis of the effect of pairs of input transformations using a comparative-based design. The transformed and optimized inputs obtained with the optimization method (see Section 2.2) are presented using a dual grid design. Each axis of the grid represents one of the transformations studied. Each cell represents a parameter combination of the transformations that have been applied to the input.

We add a subgroup level analysis as indicated in the task analysis (see Section 3) for better generalization of the instance level and a more in-depth analysis of the global level offered by ProactiV. We adopt the global analysis from ProactiV, including summarized dimensionality reduction plots per instance and per transformation for the selected distance/metric and overall performance visualization. The subgroup analysis includes aggregated visualizations for selected subgroups of instances (see Fig. 3). Next, we present the design choices on image characteristics developed to tailor ProactiV for medical images.

4.1 Image Characteristics

The design is based on the user interest in exploring image changes when analyzing input-output patterns. The user is meant to identify patterns by using domain knowledge. In contrast to regular scene images, plotting medical images directly does not reveal patterns. The images look very similar, thus, relevant changes are not perceivable. SEG-RobustEye enhances differences that the model is potentially sensitive to (see Fig 3). The relevant characteristics have been obtained through discussions with domain experts (see Section 3).

Color Characteristics: Color characteristics refer to the hue, saturation, and luminance (HSL). The HSL differences can reveal transformations in color tone and brightness between the transformed and optimized images. Visualizing these characteristics as three histograms in small grid components is ineffective, given the scale of the cells in the grids. To allow for the exploration of all three dimensions at the same time, we juxtapose the HSL characteristics in the form of a Tile Plot-based visualization (see Fig. 3.C). Each of the three dimensions is represented as a horizontal bar. A specific hue is used per each channel, orange for hue, green for saturation, and purple for luminance. The saturation of the hue encodes the number of values present in each bin. This solution presents the overall distribution of the image color dimensions, and enables visualizing the properties at the same time. To enhance the contrast, we explicitly encode differences through computing distribution distance using Kullback–Leibler divergence.

Edge Characteristics: Boundaries, or edges, are important characteristics for segmentation methods. When edges are blurred, segmentation algorithms struggle to delineate boundaries between different tissue types, leading to misclassification or incomplete segmentation. We extract the gradient information as an indication of edges. We propose to present the edge information in three ways: as an overlay (see Fig. 3.D) to contextualize, histogram of the gradient magnitude, and orientation distribution (see Fig. 4.C)

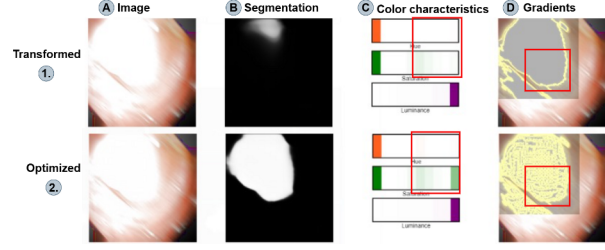


Figure 3: Transformed (1) vs Optimized (2) images. (A) The image. (B) The segmented output. (C) The HSL color characteristics. (D) The gradients overlay

for overall distribution. In Gradient orientation histograms, each bin represents an orientation, and the contribution of each gradient is weighted by its magnitude. Changes in the orientation distribution might indicate a bias towards image orientation expectations.

Frequency-based Characteristics: The frequency domain characteristics are relevant to understand the image from another view point beyond spatial. We visualize the magnitude spectrum (see Fig. 4.D). To ease comparison, we also measure the overall explicit differences in terms of magnitude and phase between transformed and optimized images. In a magnitude spectrum, the lower frequencies are in the center.

To facilitate comparison between the transformed and optimized images, we also explicitly encode pixel-wise differences and compute standard image metrics such as SSIM, PSNR, and MSE. Users can also explore overlays of the outputs and output differences between transformed and optimized instances.

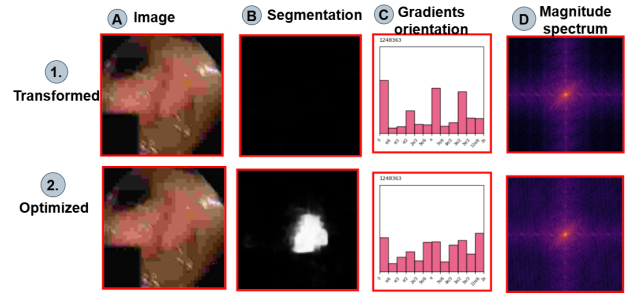


Figure 4: Transformed (1) vs Optimized (2) images. (A) The image. (B) The segmented output. (C) The gradients orientation histogram. (D) The magnitude spectrum

4.2 Subgroup analysis

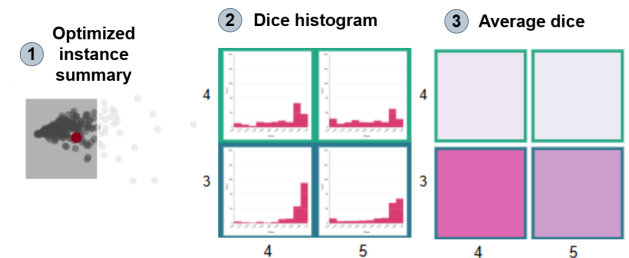


Figure 5: The subgroup analysis. The gradient metrics for the selected subgroup from the Optimized instance summary (1) are aggregated as either histograms (2) or as heatmaps (3).

For each set of transformations, the user can notice the existence of patterns in performance behavior when considering sev-

eral instances in the dataset. Using the Instance view to infer such trends demands cognitive load, as users have to rely on short memory for comparison. To easily explore group-specific behaviors and trends, the information on subgroups is summarized using quantifiable metrics and presenting the corresponding histograms. The user can select a subgroup of interesting instances in the Optimized instance summary (see Fig. 5.1), and explore the model performance, changes in HSL characteristics, gradient information, and frequency domain change information for the chosen subset of instances. The grid layout is adapted and each grid cell represents a summary of the selected subgroup of instances. Multiple metrics can be used, and we provide an option to explore them as a distribution across all instances in the selected subgroup using the histograms (see Fig. 5.2), or as a grid heatmap of the average value (see Fig. 5.3).

5 USE CASE

We present a use case developed with domain experts. The case illustrates the use of SEG-RobustEye. In this case, we look at the effect of Motion blur and Overexposure as the transformations.

SEG-RobustEye evaluates model performance on transformed and optimized datasets. Looking at the performance distribution (see Fig. 1B) it is noticed that many transformed images have low dice scores while optimized ones score much higher. This shows significant improvement after optimization. In the Optimized instance summary (see Fig. 1A), we select a subgroup of instances that behaves similarly after optimization. In this way, we filter out outliers. We further explore how this subgroup behaves by switching to the subgroup view (see Fig. 1C). We identify that the Motion blur at level 3 or higher impacts the model performance. Optimization shifts these cases from low to high dice scores, a trend seen in the dice distribution.

We see an overall increase in the edge information in the highlighted transformation levels of the subgroup (see Fig. 1D), especially as the Overexposure level increases level 8 or higher. The increase in edge information in the optimized image around the polyp (see the longer edge on the bottom left of the highlighted area in Fig. 1F3) provides an indication that the model uses the edges to perform better on the optimized images compared to the transformed ones.

We move to the instance view for further detail analysis (see Fig. 1E and Fig. 1F). For the transformed images, the model does not segment well the polyps after level three of Motion blur on specific instances (see Fig. 1E2). In some cases the polyp is under-segmented, or fully missing. For levels lower than three of Motion blur and high levels of Overexposure the polyp is over-segmented or not detected. The results improve for the optimized images (see Fig. 1E2 and Fig. 1F2).

Inspecting multiple instances in the Instance view confirms the Subgroup observations: strong edges are crucial for good performance. Our visualizations highlight increased edge magnitudes in optimized images, with stronger contours around polyps (see Fig. 1E3 and Fig. 1F3).

Frequency characteristics show that Motion blur removes high frequencies as expected, but optimization restores some, suggesting their importance for the model segmentation performance (see Fig. 1E4 and Fig. 1F4). At high Overexposure levels with strong blur, optimized images show added edge structures around overexposed areas, suggesting the model compensates for missing polyp information by adding gradient details (see Fig. 1E5 and Fig. 1F5).

6 USER STUDY

We evaluate qualitatively SEG-RobustEye by performing a user study. Three researchers (U1-U3) participated in this study. All users work in their research with endoscopy data and segmentation or classification models. They are aware of the particular input

corruptions that can occur and affect the model. One user (U1) is one of the co-authors and interacted with the system before, as they provided feedback during the development process. The other two users (U2,U3) were familiar with the background and motivation behind building such a system. We use the Kvasir-SEG dataset (See Section 2.3). The participants are guided to explore predefined transformation combinations for simplicity, but are also encouraged at the end to freely explore the system with other combinations.

The user study session consisted of three parts: an introduction to SEG-RobustEye, a set of predefined assignments followed by a free exploration, and answering a questionnaire. The introduction of the system is done in a joint session in person with all users, while the assignments and the free exploration are done in individual sessions. The assignments showcase how the proposed methods can help the user discover patterns in the input image characteristics. All the sessions were recorded and participants were asked to narrate their thoughts and experiences during the assignments. The evaluation step is individual as well, with each user providing their feedback through a mix of open questions and a fixed questionnaire (see Supplemental Material 1 for more details and results).

The users completed the assignments in the allocated time without any guidance or instructions on how to approach them (see Supplemental Material 1 for more details). Users (U1, U3) adapted to the interface quicker than U2. Measuring the time used for the introductory phase of the user study, all were able to find ways to tackle the assignments. They overall considered the proposed framework useful and could see themselves introducing this tool in their model development process.

Particularly, the instance analysis received positive feedback (U1,U2). The methods proposed for the exploration of image feature characteristics and comparison visualizations were considered "intuitive ways to approach the challenge of exploring changes in endoscopy images" (U1-U3). It was noted that, because of "the nature of the segmentation model, evaluating the color characteristics method was more challenging as there were not enough consistent patterns to be observed" (U1,U3). However, two of the users agreed that "this is an intuitive way to represent color characteristics" (U2,U3), and all the users agreed that "analyzing color characteristics is relevant". Most users also agreed that, "while useful, discovering patterns in the gradient orientation histograms can be challenging" (U1,U2). However, one user noted "challenges in generalizing patterns without the Subgroup view", which was seen as a valuable complement (U3).

Users suggested improvements, including an "automatic robustness score" (U3), "better navigation using larger visual representations" (U2), and "extending the system to support classification tasks" (U1) and "model comparisons" (U3).

7 CONCLUSION AND FUTURE WORK

SEG-RobustEye supports the global, local and subgroup level analysis of the robustness of endoscopy segmentation models by tailoring and expanding the framework proposed in ProactiV [13].

The user evaluation showed that researchers consider SEG-RobustEye particularly useful for evaluating and discovering areas of concern. However, the optimization method could be improved adding domain-specific constraints that preserve the structural components of the transformed image to the optimized one. To ensure the scalability of the framework for large datasets the current implementation should be extended such that preprocessing is not needed. Finally, we used endoscopy application as a test-bed but evaluation beyond the endoscopy scenario, and with more users would be interesting future work.

Overall, SEG-RobustEye is a visual analytics system based on ProactiV that provides a way in which researchers can interactively visualize model behavior and image characteristics to identify and understand breaking points, at instance, subgroup and global levels.

SUPPLEMENTAL MATERIALS STRUCTURE

In the Supplemental Materials, we included further details and results of the user study (Supplemental Material 1, Section 1), a second use case of our system (Supplemental Material 1, Section 2), an overview screenshot of the whole system and example images of the Kvasit-SEG dataset and used image corruptions (Supplemental Material 2), the survey used to gather user feedback, including the survey results (Supplemental Material 3), and a video explaining the method and an use case (Supplemental Material 4).

REFERENCES

- [1] S. F. Dodge and L. J. Karam. Understanding how image quality affects deep neural networks. In *2016 Eighth International Conference on Quality of Multimedia Experience (QoMEX)*, pp. 1–6, 2016. 1
- [2] S. F. Dodge and L. J. Karam. A study and comparison of human and deep learning recognition performance under visual distortions. In *2017 26th International Conference on Computer Communication and Networks (ICCCN)*, pp. 1–7, 2017. 1
- [3] K. Faust, Q. Xie, D. Han, K. Goyle, Z. Volynskaya, U. Djuric, and P. Diamandis. Visualizing histopathologic deep learning classification and anomaly detection using nonlinear feature space dimensionality reduction. *BMC bioinformatics*, 19:1–15, 2018. 2
- [4] D. Hendrycks and T. Dietterich. Benchmarking neural network robustness to common corruptions and perturbations. In *International Conference on Learning Representations*, 2019. 1
- [5] T. J. M. Jaspers, T. G. W. Boers, C. H. J. Kusters, M. R. Jong, J. B. Jukema, A. J. de Groof, J. J. Bergman, P. H. N. de With, and F. van der Sommen. Investigating the impact of image quality on endoscopic ai model performance. In S. Wu, B. Shabestari, and L. Xing, eds., *Applications of Medical Artificial Intelligence*, pp. 32–41. Springer Nature Switzerland, 2024. 1
- [6] T. J. M. Jaspers, T. G. W. Boers, C. H. J. Kusters, M. R. Jong, J. B. Jukema, A. J. de Groof, J. J. Bergman, P. H. N. de With, and F. van der Sommen. Robustness evaluation of deep neural networks for endoscopic image analysis: Insights and strategies. *Medical Image Analysis*, 94:103157, 2024. 1, 2
- [7] D. Jha, P. H. Smedsrud, M. A. Riegler, P. Halvorsen, T. de Lange, D. Johansen, and H. D. Johansen. Kvasir-seg: A segmented polyp dataset. In Y. M. Ro, W.-H. Cheng, J. Kim, W.-T. Chu, P. Cui, J.-W. Choi, M.-C. Hu, and W. De Neve, eds., *MultiMedia Modeling*, pp. 451–462. Springer International Publishing, 2020. 2
- [8] S. Karahan, M. Kilinc Yildirim, K. Kirtac, F. S. Rende, G. Butun, and H. K. Ekenel. How image degradations affect deep cnn-based face recognition? In *2016 International Conference of the Biometrics Special Interest Group (BIOSIG)*, pp. 1–5. IEEE, 2016. 1
- [9] C. H. J. Kusters, T. G. W. Boers, T. J. M. Jaspers, J. B. Jukema, M. R. Jong, K. N. Fockens, A. J. de Groof, J. J. Bergman, F. van der Sommen, and P. H. N. de With. Cnns vs. transformers: Performance and robustness in endoscopic image analysis. In S. Wu, B. Shabestari, and L. Xing, eds., *Applications of Medical Artificial Intelligence*, pp. 21–31. Springer Nature Switzerland, 2024. 1, 2
- [10] H. X. Li, S. Jorgensen, J. Holodnak, and A. B. Wollaber. Scatteruq: Interactive uncertainty visualizations for multiclass deep learning problems. In *2023 IEEE Visualization and Visual Analytics (VIS)*, pp. 246–250. IEEE, 2023. 2
- [11] C. Park, S. Yang, I. Na, S. Chung, S. Shin, B. C. Kwon, D. Park, and J. Choo. Vatun: Visual analytics for testing and understanding convolutional neural networks. In *EuroVis (Short Papers)*, pp. 7–11, 2021. 2
- [12] Y. Pei, Y. Huang, Q. Zou, X. Zhang, and S. Wang. Effects of image degradation and degradation removal to cnn-based image classification. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 43(4):1239–1253, 2021. 1
- [13] V. Prasad, R. J. van Sloun, A. Vilanova, and N. Pezzotti. Proactiv: Studying deep learning model behavior under input transformations. *IEEE Transactions on Visualization and Computer Graphics*, 30(8):5651–5665, 2024. 1, 2, 3, 4
- [14] C. Severn, K. Suresh, C. Görg, Y. S. Choi, R. Jain, and D. Ghosh. A pipeline for the implementation and visualization of explainable machine learning for medical imaging using radiomics features. *Sensors*, 22(14):5205, 2022. 2
- [15] S. Sietzen, M. Lechner, J. Borowski, R. Hasani, and M. Waldner. Interactive analysis of cnn robustness. In *Computer Graphics Forum*, vol. 40, pp. 253–264. Wiley Online Library, 2021. 2
- [16] J. Wang, L. Gou, W. Zhang, H. Yang, and H.-W. Shen. Deepvid: Deep visual interpretation and diagnosis for image classifiers via knowledge distillation. *IEEE transactions on visualization and computer graphics*, 25(6):2168–2180, 2019. 2
- [17] M. Xiao, L. Zhang, W. Shi, J. Liu, W. He, and Z. Jiang. A visualization method based on the grad-cam for medical image segmentation model. In *2021 International Conference on Electronic Information Engineering and Computer Science (EIECS)*, pp. 242–247. IEEE, 2021. 2